3/21/22

Monday, March 21, 2022 11:20 AM

M168_Notes_7.2

7.2 Definition of Probability

The probability of an event, $P(E)$ is a number between 0 and 1, inclusive. If $P(E)=0$, then the event E is impossible. If $P(E)=1$, then the event E is certain.

The ***theoretical probability*** of an event E occurring is based on the sample space S having equally likely outcomes. Then probability of the event E occurring is

$$P(E)=\frac{\text{number of outcomes in event E}}{\text{number of outcomes in the sample space}}=\frac{n(E)}{n(S)}$$

Example

Consider flipping a fair coin three times. Find the following probabilities:

(a) Exactly one head is seen. $n(E)=3$, $n(S)=2^3=8$

$P(E)=\frac{3}{8}$

(b) Two or more heads are seen. $1/2$

(c) More than 3 heads are seen. 0

We can also calculate the ***empirical probability*** of an event by doing an experiment many times. Roll a six sided die and count the number of times a 1 is observed.

# of tosses (m) rolls	# of 1's rolled (n)	relative frequency (n/m)
30	4	4/30
100	16	16/100

Consider the uniform sample space $S=\{s_1,s_2, ...,s_n\}$, with *n* outcomes. The *n* events that contain a single outcome, $\{s_1\}$, $\{s_2\}$ . . . $\{s_n\}$ are called ***simple*** events.

Events that can't occur at the same time are called ***mutually exclusive***. Note that the simple events are mutually exclusive.

E is the event that the sum of the numbers shown ~~uppermost~~ is 7
F is the event that the red die shows a 1
G is the event that the green die shows a 6
H is the event that the sum of the numbers shown ~~uppermost~~ is 10
Which events are mutually exclusive?

$E \cap F = \{1 \sim 6\}$

$F \cap G = \{1 \sim 6\}$

$E \cap H = \phi$

$\boxed{F \cap H = \phi}$

1~1 2~1 3~1 4~1 5~1 6~1
1~2 2~2 3~2 4~2 5~2 6~2
1~3 2~3 3~3 4~3 5~3 6~3
1~4 2~4 3~4 4~4 5~4 6~4
1~5 2~5 3~5 4~5 5~5 6~5
1~6 2~6 3~6 4~6 5~6 6~6

A ***probability distribution table*** has the following properties:

1. Each of the entries is mutually exclusive with all other entries
2. The sum of the probabilities is 1

PROBABILITY DISTRIBUTION TABLE:

Event	probability
E_1	p_1
E_2	p_2
E_3	p_3

$E_1 \cap E_2 = \phi$

$E_2 \cap E_3 = \phi$

$E_1 \cap E_3 = \phi$

$p_1 + p_2 + p_3 = 1$

Example

Find the probability distribution table for the number of heads when a coin is tossed 3 times.

Event	Probability
0 Heads	1/8
1 Head	3/8
2 Heads	3/8
3 Heads	1/8

0 - Heads $= E_0 = \{TTT\}$

1 - Head $= E_1 = \{HTT, THT, TTH\}$

What is the probability of 2 or more heads?

1/2

Suppose the instructor of a class polled the students about the number of hours spent per week studying math during the previous week. The results were 69 students studied two hours or less, 128 students studied more than two hours but 4 or less hours, 68 students studied more than 4 hours but less than or equal to 6 hours, 30 students studied more than 6 hours but less than or equal to 8 hours and 14 students studied more than 8 hours.

Arrange this information into a PDT and find the probability that a student studied more than 4 hours per week

Hours spent	# of students	Probability
$0 < t \leq 2$	69	69/309
$2 < t \leq 4$	128	128/309
→ $4 < t \leq 6$	68	68/309
→ $6 < t \leq 8$	30	30/309
→ $8 < t$	14	14/309

$$P(E_{>4}) = \frac{68 + 30 + 14}{309} = \frac{112}{309}$$

What is the probability of rolling a sum 2 or a sum of 12 using two fair die?

1~1	2~1	3~1	4~1	5~1	6~1
1~2	2~2	3~2	4~2	5~2	6~2
1~3	2~3	3~3	4~3	5~3	6~3
1~4	2~4	3~4	4~4	5~4	6~4
1~5	2~5	3~5	4~5	5~5	6~5
1~6	2~6	3~6	4~6	5~6	6~6

$$\frac{2}{36} = \frac{1}{18}$$

What is the probability of rolling a sum of 7?

1~1	2~1	3~1	4~1	5~1	6~1
1~2	2~2	3~2	4~2	5~2	6~2
1~3	2~3	3~3	4~3	5~3	6~3
1~4	2~4	3~4	4~4	5~4	6~4
1~5	2~5	3~5	4~5	5~5	6~5
1~6	2~6	3~6	4~6	5~6	6~6

$$\frac{6}{36} = \frac{1}{6}$$